

Morley's Trisector Theorem: A Multi-perspective approach

A Report

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by

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CERTIFICATE

Student's Declaration

I hereby certify that the work presented in this report titled "**Morley's Trisector Theorem: A Multi-perspective approach**" is carried out by me. This is a new presentation of a work existing in the literature. The contents that are taken from the literature are cited properly by providing the best possible reference. It is further certified that the use of AI is limited to simulation and analysis of data. All the other contents were prepared by me.

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ABSTRACT

Morley's Trisector Theorem is a fundamental result in elementary geometry stating that the three points of intersection of the adjacent trisectors of any triangle's angles form an equilateral triangle. Discovered by Frank Morley in 1899, the theorem has since received multiple independent proofs. This report presents five proofs drawn from different mathematical domains (trigonometry, complex-number algebra, synthetic geometry, group theory, and circle geometry).

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1. INTRODUCTION

Let $\triangle ABC$ be an arbitrary triangle. If we trisect each interior angle of the triangle and take the intersections of the adjacent trisectors, we obtain three interior points.

1.1. The Theorem.

Theorem 1.1 (Morley, 1899 [5]). *For any triangle $\triangle ABC$, the pairwise intersections of the adjacent angle trisectors form the vertices of an equilateral triangle. This inner equilateral triangle is called the Morley triangle.*

Notation. Throughout this paper, we denote the interior angles of $\triangle ABC$ as $m(\angle A) = 3\alpha$, $m(\angle B) = 3\beta$, and $m(\angle C) = 3\gamma$. Consequently, we have the relation $\alpha + \beta + \gamma = \pi/3$. The vertices of the Morley triangle will be denoted P, Q, R , where P is near side BC , Q near AC , and R near AB .

2. PROOF 1: TRIGONOMETRY

This approach, variations of which are found in Newman [7] and Taylor and Marr [9], computes the side lengths of the Morley triangle to demonstrate their equality.

2.1. Setting up the Geometry. We normalize the circumcircle of the primary triangle $\triangle ABC$ such that its circumradius is $R = 1$. By the extended Law of Sines, a chord subtending an inscribed angle θ has length $2R \sin(\theta)$. The side lengths of the triangle can be expressed as $BC = 2 \sin(3\alpha)$, $AC = 2 \sin(3\beta)$, and $AB = 2 \sin(3\gamma)$.

2.2. Computing Intermediary Sides. Consider $\triangle BPC$. The angles at B and C generated by the adjacent trisectors are β and γ , respectively. The remaining angle is $m(\angle BPC) = \pi - (\beta + \gamma)$. Applying the Law of Sines to $\triangle BPC$ to find the length of BP , we obtain:

$$BP = \frac{BC \sin(\gamma)}{\sin(\pi - (\beta + \gamma))} = \frac{2 \sin(3\alpha) \sin(\gamma)}{\sin(\pi/3 + \alpha)}$$

The denominator simplifies because $\beta + \gamma = \pi/3 - \alpha$, giving $\pi - (\pi/3 - \alpha) = 2\pi/3 + \alpha$. The sine of this angle is identically $\sin(\pi/3 - \alpha)$.

We invoke the trigonometric triple-angle identity $\sin(3\alpha) = 3 \sin(\alpha) - 4 \sin^3(\alpha)$, which factors as:

$$\sin(3\alpha) = 4 \sin(\alpha) \sin\left(\frac{\pi}{3} + \alpha\right) \sin\left(\frac{\pi}{3} - \alpha\right)$$

Substituting this expansion into our equation for BP , the $\sin(\pi/3 - \alpha)$ term in the numerator cancels with the denominator, leaving:

$$BP = 8 \sin(\alpha) \sin\left(\frac{\pi}{3} + \alpha\right) \sin(\gamma)$$

By an identical argument applied to $\triangle BRA$ on the adjacent side of the triangle, we obtain:

$$BR = 8 \sin(\gamma) \sin\left(\frac{\pi}{3} + \gamma\right) \sin(\alpha)$$

2.3. Evaluating the Morley Triangle. We now evaluate the side PR of the Morley triangle. Consider $\triangle BPR$, where the central trisected angle is $m(\angle PBR) = \beta$. The Law of Cosines dictates:

$$PR^2 = BP^2 + BR^2 - 2(BP)(BR) \cos(\beta)$$

Substituting our expressions for BP and BR , we factor out the common terms $64 \sin^2(\alpha) \sin^2(\gamma)$:

$$PR^2 = 64 \sin^2(\alpha) \sin^2(\gamma) \left[\sin^2\left(\frac{\pi}{3} + \alpha\right) + \sin^2\left(\frac{\pi}{3} + \gamma\right) - 2 \sin\left(\frac{\pi}{3} + \alpha\right) \sin\left(\frac{\pi}{3} + \gamma\right) \cos(\beta) \right]$$

To simplify the bracketed expression, consider an auxiliary triangle inscribed in a unit circle with angles $\frac{\pi}{3} + \alpha$, $\frac{\pi}{3} + \gamma$, and β (these sum to π since $\alpha + \beta + \gamma = \pi/3$). By the Law of Cosines applied to this auxiliary triangle, the bracketed term represents the square of the chord opposite the angle β , which is $\sin^2(\beta)$.

Taking the square root yields:

$$PR = 8 \sin(\alpha) \sin(\beta) \sin(\gamma)$$

Because this distance formula is invariant under permutations of α, β , and γ , analogous calculations for PQ and QR yield identical lengths. Therefore, $\triangle PQR$ is an equilateral triangle.

3. PROOF 2: COMPLEX NUMBERS

This algebraic proof, formalized by Letac and discussed by Cundy [3], maps the problem onto the complex unit circle. It relies on points represented as unit-modulus complex numbers (turns).

3.1. Embedding in the Unit Circle. Let the circumcircle of $\triangle ABC$ be the unit circle $U \subset \mathbb{C}$. We define the affixes of vertices A, B, C as a^3, b^3, c^3 , where $a, b, c \in U$. The points of trisection of the consecutive arcs AB, BC, CA are given by the intermediate turns a^2b, ab^2 ; b^2c, bc^2 ; $\omega ac^2, \omega^2 a^2c$, where $\omega = e^{2\pi i/3}$.

3.2. Chord Intersections. The equation of a line passing through two turns $t, u \in U$ is given by $z + tuz^* = t + u$, where z^* is the complex conjugate. The Morley vertex x is the intersection of two interior trisector chords. Substituting the turns of the trisected arcs yields a simultaneous system:

$$\begin{aligned} x + ab^2c^3x^* &= ab^2 + c^3 \\ x + \omega ab^3c^2x^* &= \omega ac^2 + b^3 \end{aligned}$$

Subtracting the second equation from the first and dividing the result by $(c - \omega b)$ yields the conjugate affix of the first Morley vertex:

$$a^2b^2c^2x^* = \omega^2(ab^2 - a^2b) + ac^2 - \omega a^2c + \omega abc$$

Analogous algebraic expressions are derived for y^* and z^* by cyclic permutation $a \mapsto b \mapsto c \mapsto a$.

To prove $\triangle xyz$ is equilateral, we verify the condition $x + \omega y + \omega^2 z = 0$. Summing our expressions for $x^*, y^*,$ and z^* scaled by the appropriate powers of ω results in the cancellation of the polynomial terms:

$$x^* + \omega^2 y^* + \omega z^* = 0$$

Taking the complex conjugate of this equation yields $x + \omega y + \omega^2 z = 0$. This linear relation shows that the vectors connecting the points x, y, z correspond to rotations by $\pi/3$, indicating they form an equilateral triangle.

4. PROOF 3: SYNTHETIC GEOMETRY

John Conway's synthetic proof [2] begins with the inner Morley triangle and constructs a larger triangle similar to $\triangle ABC$ around it.

4.1. The Puzzle Pieces. Let $\triangle PQR$ be an equilateral triangle with side length s . For any angle θ , we define $\theta^+ = \theta + \pi/3$ and $\theta^{++} = \theta + 2\pi/3$. Let $\alpha, \beta, \gamma > 0$ be chosen such that $\alpha + \beta + \gamma = \pi/3$.

We construct six auxiliary triangles around the central $\triangle PQR$ (three abutting triangles and three spoke triangles). The abutting triangle corresponding to vertex A is assigned angles $\alpha, \beta^+, \gamma^+$. The spoke triangle corresponding to side BC is assigned angles $\gamma^{++}, \alpha, \beta$. We scale these six triangles to force the edges meeting the central equilateral $\triangle PQR$ to be of length s .

4.2. Angle Assembly. We assemble these seven pieces. For the geometry to form a closed planar polygon, the sum of the angles around each internal vertex must be 2π . At a typical internal vertex, the contributing angles are α^+ (spoke), β^{++} (adjacent spoke), γ^+ (abutting), and $\pi/3$ (central equilateral $\triangle PQR$). Summing these yields:

$$(\alpha + \pi/3) + (\beta + 2\pi/3) + (\gamma + \pi/3) + \pi/3 = \alpha + \beta + \gamma + \frac{5\pi}{3}$$

Given the condition $\alpha + \beta + \gamma = \pi/3$, this sum is 2π . By construction, the vertices of the central equilateral piece lie on the concurrent trisectors of the assembled outer triangle. The constructed outer triangle has angles $3\alpha, 3\beta, 3\gamma$ and is similar to $\triangle ABC$, proving the theorem.

5. PROOF 4: GROUP THEORY

Alain Connes formulated a proof [1] characterizing Morley's theorem as an algebraic property of the affine group of the complex line.

5.1. The Affine Group. Let G be the affine group of $\mathbb{C}$. An element $g \in G$ takes the form $g(x) = ax + b$, where $a \in \mathbb{C}^*$ and $b \in \mathbb{C}$. For any $a \neq 1$, g possesses a fixed point $\text{fix}(g) = \frac{b}{1-a}$.

5.2. Triangle Encoding. Let g_1, g_2, g_3 be affine rotations about the vertices A, B, C by angles $2\alpha, 2\beta, 2\gamma$ respectively. The multiplicative parameters satisfy $a_1 a_2 a_3 = j$, where $j = e^{2\pi i/3}$. The adjacent trisector intersections of the Morley triangle are the fixed points of the pairwise products of these affine maps: $\alpha = \text{fix}(g_1 g_2)$, $\beta = \text{fix}(g_2 g_3)$, and $\gamma = \text{fix}(g_3 g_1)$.

5.3. The Equilateral Condition. Because the interior angles of a Euclidean triangle sum to π , the product of the cubes of these transformations must equal the identity transformation: $g_1^3 g_2^3 g_3^3 = 1$.

This group identity implies that its translational part, b , is zero. Connes expands the translational part into a product of polynomials:

$$b = -ja_1^2 a_2(a_1 - j)(a_2 - j)(a_3 - j)(\alpha + j\beta + j^2\gamma)$$

Since $g_1^3 g_2^3 g_3^3 = 1$ implies $b = 0$, and the pairwise products are rotations ($a_k - j \neq 0$), the final factor is zero:

$$\alpha + j\beta + j^2\gamma = 0$$

This relation is the standard algebraic condition for α, β, γ to form an equilateral triangle.

6. PROOF 5: CIRCLE GEOMETRY

This proof utilizes inscribed angles and auxiliary circles as formulated by Nancy Walls [10].

6.1. Construction. Start with an equilateral triangle $P_1P_2P_3$ and three angles a_1, a_2, a_3 such that their sum is $\pi/3$. Construct three auxiliary circles S_i ($i = 1, 2, 3$) passing through P_j, P_k such that the chord P_jP_k subtends an angle a_i at the circumference.

Identify points Q_i, R_i on the boundary of S_i such that the chords Q_iP_k and P_jR_i are equal in length to the base chord P_jP_k . Extend the lines Q_kR_i and R_jQ_i until they intersect at the outer vertices X_i .

6.2. Angle Chase. An angle chase demonstrates that the pentagon $P_jQ_kX_iR_jP_k$ has interior angles summing to 3π . This implies $m(\angle Q_iX_iR_i) = 3a_i$. Because $3a_i$ is the angle subtended at the circumference of S_i , the point X_i lies on the circle S_i .

Since the chords are equal in length, the inscribed angles they subtend are equal. Therefore, the lines X_iP_j and X_iP_k trisect the angle $\angle X_i$. Consequently, $X_1X_2X_3$ forms the outer triangle similar to $\triangle ABC$, with $P_1P_2P_3$ as its inner Morley triangle.

7. BEYOND THE PROOFS

7.1. Cardioids and Inversive Geometry. Frank Morley's initial discovery was connected to algebraic curves, specifically inscribed cardioids [6]. Morley noted that the 27 intersection points of the general trisectors of $\triangle ABC$ align with the centers of 27 doubly inscribed cardioids, situating the theorem within inversive geometry. Oakley and Baker [8] provide historical context regarding Morley's primary focus on cardioid systems rather than the triangle theorem directly.

7.2. Holonomy. The theorem can be analyzed via differential geometry. Utilizing a method formalized by Doyle and Sethi [4], one can construct marked similarity triangles (MSTs) around a vertex to compute a local invariant known as *holonomy* (the scale factor between a leading edge and a lagging edge). Demonstrating that the product of these local holonomies over a fundamental period equates to 1 proves that the inner geometry maps onto the Euclidean plane without distortion. This invariant serves as the basis for Conway's puzzle-piece construction [2].

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